**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 029 · Karnaugh maps 2: simplify to gates

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. In a Karnaugh-map pair, A stays 0 and B changes between 0 and 1. What is the simplified term?

Q2. Show the array after one complete left-to-right pass of ascending bubble sort on [8, 3, 6, 2] .

Q3. C#: int[,,] scores = new int[2, 3, 4]; How many values can it store? What does scores.GetLength(2) return?

Q4. Add 01000000 0011 and 01000000 0010 . Format: 8-bit two’s-complement fractional mantissa + 4-bit two’s-complement exponent . Show exponent alignment and the final normalised result.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

Recall the term — efficient group

Recall the term — overlap

Recall the term — wraparound

Recall the term — literal

Recall the term — product term

Recall the term — simplified expression

Recall the term — equivalent

### New learning

Find the 1s — Look at the 2×2 map on the left. Three squares contain 1 . We need to cover them in groups of neighbouring 1s. — Use rectangles containing 1, 2, 4, 8 or 16 cells. Include only 1s, and choose the largest useful groups. Diagonal cells are not neighbours.

A square may belong to two groups — This is overlap . It is allowed when it makes the expression simpler.

Keep the inputs that stay the same — Bottom pair: A stays 1; B changes. Keep A. — Right pair: B stays 1; A changes. Keep B.

The outside columns form one group — BC runs 00, 01, 11, 10 . The first and last columns are neighbours: only B changes between 00 and 10 . — Imagine rolling the map so its left and right edges meet. The four 1s make a 2 × 2 group ; the middle 0s are outside it.

Compare all four cells — A changes between the two rows → remove A. — B changes between columns 00 and 10 → remove B.

### Worked example

How do two groups become a circuit?

1. Find the two pairs of 1s — ORANGE · TOP: circle columns 01 and 11 where A = 0. — BLUE · BOTTOM: circle columns 11 and 10 where A = 1. — Follow the same coloured groups on the map on the left. Each pair becomes one term.

2. Keep ¬A and C — A = 0 in both squares, so write ¬A . C = 1 in both, so keep C . B changes , so remove B.





### Exit Ticket · Model answer

True. Simplification must preserve the Boolean output for every input combination.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. ¬A A stays 0, so retain NOT A (¬A). B changes, so remove B.

Q2. [3, 6, 2, 8] Swap 8 and 3 → [3, 8, 6, 2]; swap 8 and 6 → [3, 6, 8, 2]; swap 8 and 2 → [3, 6, 2, 8]. The largest value is now last.

Q3. 24 values; GetLength(2) = 4 2 layers × 3 rows × 4 columns = 24 values. Dimension 2 is the third dimension (columns).

Q4. 01100000 0011 (denary 6) Align to exponent 3 by shifting the second mantissa right: 01000000 → 00100000. Add the mantissas: 01000000 + 00100000 = 01100000. The exponent remains 0011. This is normalised: $0.75 \times 2^3 = 6$.

### You Do pages · Independent questions

Q1. Bottom row: A stays 1; B changes. Keep A.
Right column: B stays 1; A changes. Keep B.
Join with OR: Either pair can make Q=1.
Gate: A and B feed one OR gate.

Q2. Circle: Two 1s in the top row under BC=11 and BC=10.
Compare: A stays 0, B stays 1, and C changes.
Write: Keep NOT A (¬A) and B, joined by AND. Leave C out.
Gates: NOT A, then AND with B.

Q3. One wraparound group: Both A rows at BC=00 and BC=10. The left and right edges touch.
Compare: A and B change, while C always stays 0.
Write: Remove A and B; retain ¬C.
Gate: One NOT gate with input C.

Q4. Orange, top-left group of four: AB=00/01; CD=00/01. A stays 0 and C stays 0. B and D change. First term = ¬A ∧ ¬C.
Blue, bottom wraparound group of four: AB=11/10; CD=00/10. A stays 1 and D stays 0. B and C change. Second term = A ∧ ¬D.
Join: P is 1 when either term is true, so use OR.
Gate plan: NOT A, NOT C → AND; A, NOT D → AND; then OR the two outputs. B is not needed.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

3. Keep A and B — A = 1 in both squares, so keep A . B = 1 in both, so keep B . C changes , so remove C.

4. Join the two pairs with OR — Either coloured pair makes Q = 1 . AND the inputs inside each pair; OR the two results together.

5. Draw the actual logic gates — Trace it: A goes through NOT for the orange pair; A also goes directly to the blue pair. Each pair feeds an AND gate. OR the two AND outputs to get Q.

## We Do · Practise with guidance

Find one pair

1. Find the 1s

Circle the two 1s. Which two inputs always equal 1? Which input changes?

2. Compare your circles

The pair is on row A=1, across columns BC=01 and 11.

3. Simplify the group(s)

Use the two unchanged inputs to make one Boolean term. Which gate joins them?

4. Why is this correct?

Let two groups share a 1

Find the two largest pairs. Is it okay for the top-right 1 to be used twice?

Top row and right column both contain a valid pair, sharing the top-right 1.

Top pair: A stays 0. Right pair: B stays 1. Write the two terms and join them.

Remember the outside edges touch

The BC columns are 00, 01, 11, 10. Circle the four 1s at the two outside edges as ONE group.

The outside columns (BC=00 and BC=10) are neighbours across the edge.

Across the four 1s, which variable stays 0? Which two change? Write Q.

Try a bigger Karnaugh map

Rows AB and columns CD use Gray code. Find two groups of four: top-left, then bottom-left and bottom-right across the edges.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

1. Find one pair
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.
A=0 and B=0 throughout the pair; C changes.
Gate plan: Invert A and B, then AND them.

### Question 2

2. Let two groups share a 1
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.
Top row: ¬A. Left column: ¬B. The top-left cell belongs to both groups.
Gate plan: Invert A and B, then OR them.

### Question 3

3. Wrap around the outside edges
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.
The two outside columns form ONE group of four. B=0; A and C change.
Gate plan: One NOT gate connected to B.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

### Question 4

4. Two groups in a bigger map
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression P.
Draw the logic gates and connections.
Top-right four: B=0 and C=1. Bottom outer four (across the edges): A=1 and D=0.
Gate plan: NOT B and NOT D; two AND gates feeding one OR gate.

## Exit Ticket · Answer before checking

A K-map simplification is valid only if the simplified expression gives the same output for every possible input combination.